

Bootstrapping Correlated Data¹

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INTRODUCTION

The bootstrap is a nonparametric, computer-intensive procedure that can be used to provide approximate confidence intervals for a wide class of sample statistics. Efron (1982) considers the bootstrap and its relation to other resampling procedures in the case where observations are independent and identically distributed. This note describes a procedure for applying the bootstrap when observations are correlated.

The basic idea involves transforming correlated observations to uncorrelated quantities, forming a bootstrap sample from these quantities, and transforming back to a quasibootstrap sample from the original observations. One application of this procedure is to provide confidence intervals for estimation variances, which are functionals of the autocovariance function. In related work, not involving the bootstrap, Switzer (1984) develops statistical tests for parameters of an auto-covariance function of specified form.

THE BOOTSTRAP

The bootstrap is based on the same principle as Monte Carlo simulation. Suppose we are interested in estimating the sampling distribution of a statistic, T_n , calculated from a sample, Z_n , of size n drawn from a probability distribution F . In Monte Carlo simulation, a large number of independent samples each of size n is drawn from F . For each sample, the statistic T_n is calculated. The empirical distribution of values of T_n so calculated is taken as an estimate of the sampling distribution of T_n . Note that to apply Monte Carlo simulation it is necessary to know F . Monte Carlo simulation is in this sense parametric. The

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difference between Monte Carlo simulation and the bootstrap is that it is not necessary to specify F to apply the bootstrap. Instead, the bootstrap uses the empirical distribution of the sample, Z_n , itself from which to resample.

For example, suppose the original sample contains n independent and identically distributed observations. The statistic T_n is calculated. We wish to estimate the sampling distribution of T_n . That is, we wish to know how different T_n might have been had a different sample of size n been drawn from the same underlying probability distribution. Each bootstrap sample is chosen by sampling n values at random with replacement from Z_n . As before, the statistic T_n is calculated and the procedure is repeated many times. The empirical distribution of calculated values of T_n is taken as an estimate of the sampling distribution of T_n . In some cases, smoothing the empirical distribution of observations—by, say, passing a kernel over it—prior to performing bootstrap has proven useful.

Suppose now that elements of Z_n represent a sample from a realization of a second-order stationary random process, $Z(x)$, where x is location in Euclidean space of arbitrary dimension. Suppose further that the statistic of interest depends on the pattern of autocorrelation in the process as expressed in the autocovariance function, $C(h)$, where h is displacement in space. An example of such a statistic is the estimation variance associated with a fixed data configuration (Journel and Huijbregts, 1978, p. 53). The bootstrap procedure described above will not work in this case. In particular, randomizing observations without regard to their relative locations will destroy the pattern of autocorrelation exhibited by the original sample.

Suppose as above that Z_n is an n vector of observations from a realization of a second-order stationary random process, $Z(x)$, and that the covariance matrix for Z_n is C . Suppose further that

$$E(Z_n) = 0_n \quad (1)$$

where 0_n is an n vector of zeroes. In practice Z_n can be centered by subtracting an estimate of the stationary mean from each observation. Finally, suppose that the elements of Z_n are identically marginally distributed.

Because C is a positive definite matrix, it has the Cholesky decomposition

$$C = LL' \quad (2)$$

where L is lower triangular and t denotes transpose (Strang, 1976). Consider the values defined by

$$U_n = L^{-1}Z_n \quad (3)$$

The elements of U_n will have zero mean and covariance matrix equal to the identity matrix. If in addition to being uncorrelated and having common mean and variance, elements of U_n are assumed to be independently and identically

distributed, then a bootstrap sample can be drawn by sampling n values at random with replacement from the elements of U_n . This bootstrap sample, U'_n , can then be transformed

$$Z'_n = LU'_n \quad (4)$$

to form a quasibootstrap sample, Z'_n , from the original observations. Note that if elements of Z_n were centered as described above, the estimate of the mean should be added back to the elements of Z'_n .

Once Z'_n is found, a new autocovariance function, $C'(h)$, can be estimated. Using $C'(h)$, the estimation variance (or other quantity) of interest can be calculated. Repeating the procedure many times leads to an empirical distribution of estimation variances which can be taken as an estimate of the sampling distribution of the estimation variance.

In practice, the question arises as to whether using the estimated (and not the true) covariance matrix in the bootstrap is an acceptable approximation. The essence of the bootstrap—for independent or correlated values—is that it relies as much as possible on the data. Errors will occur if the data are not representative of the underlying population. How sensitive results of this bootstrap are to misspecification of the covariance matrix is an open question.

APPLICATION

For demonstration purposes, a simple application of bootstrap was carried out on a one-dimensional simulation. One hundred values from a Gaussian moving average process of order 10 were generated. The experimental autocovariance function for the simulation (Fig. 1) exhibited large-scale periodic behavior which would not be modeled in practice. Therefore, it was set to zero for all lags 10 or greater. Apart from this, no modeling of the autocovariance function was necessary.

The purpose of the study was to demonstrate how the bootstrap would be used to estimate sampling distribution of an estimation variance. The estimation variance for the data configuration shown (Fig. 2) was calculated from the experimental autocovariance function. This estimation variance may be expressed

$$s_E^2 = 1.5C(0) - 2C(1) + 0.5C(2) \quad (5)$$

(Journel and Huijbregts, 1978, p. 53). Its calculated value was 1.47.

Marginal bootstrap sampling distributions for $C(0)$, $C(1)$, and $C(2)$ based on 500 bootstrap replications are shown in (Fig. 3). Also reported are experimental values for $C(0)$, $C(1)$, and $C(2)$ based on the autocovariance function (Fig. 1). Preliminary results suggest that means of the bootstrap sampling distributions are closer to experimental values when the histogram of elements of U_n is smoothed. Presumably, this is because smoothing adds a tail to this histogram.

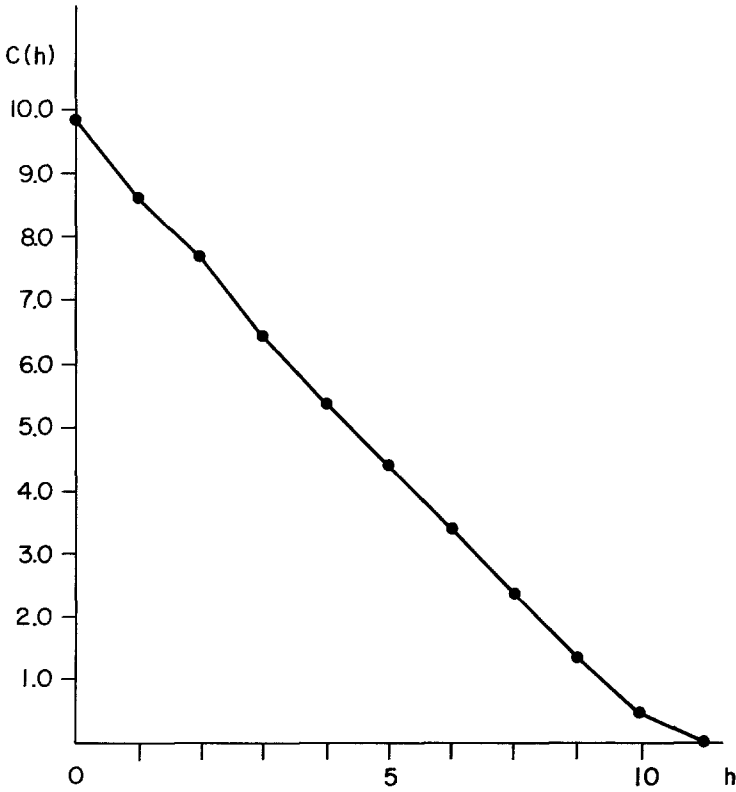


Fig. 1. Experimental autocovariance function for simulated data set.

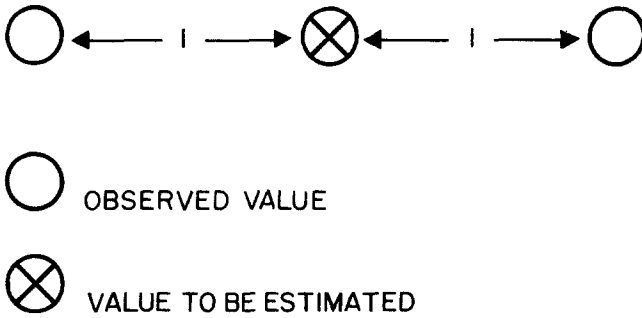


Fig. 2. Data configuration used in bootstrap application. Each datum receives a weight of 0.5.

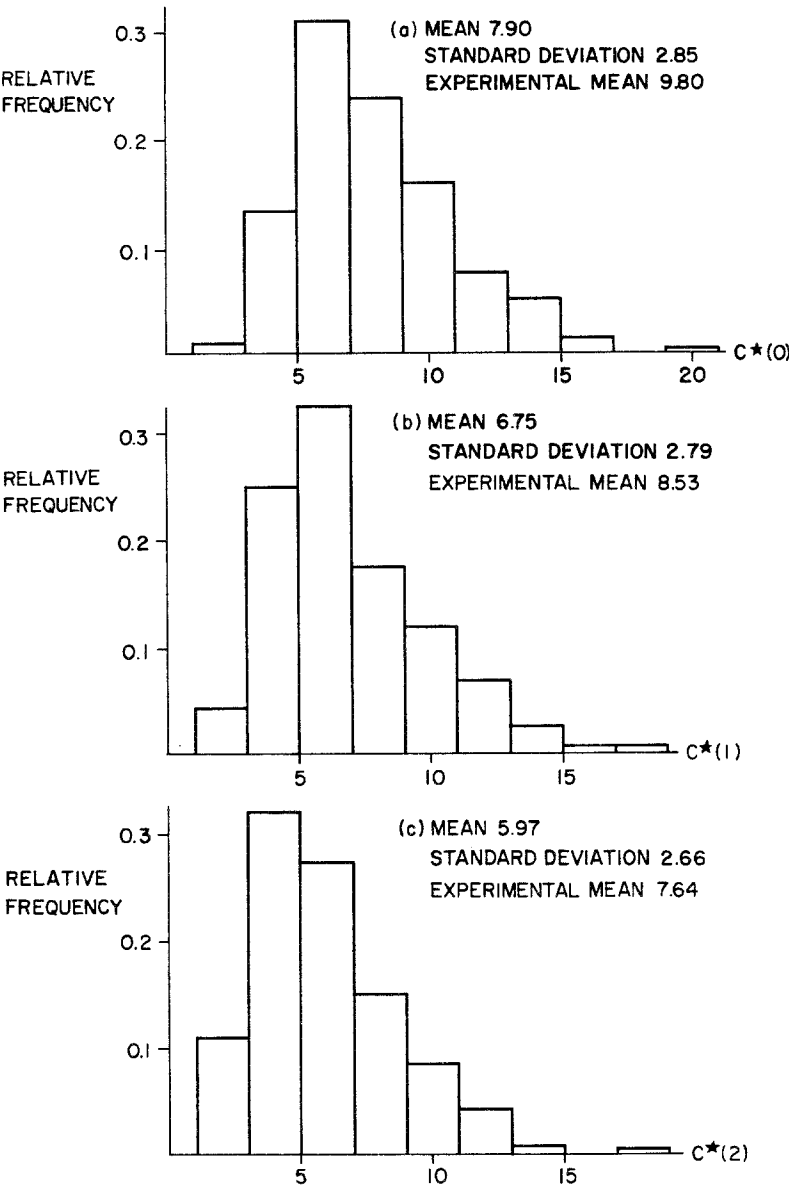


Fig. 3. Bootstrap sampling distributions for $C(0)$, $C(1)$, and $C(2)$, each based on 500 bootstrap replications.

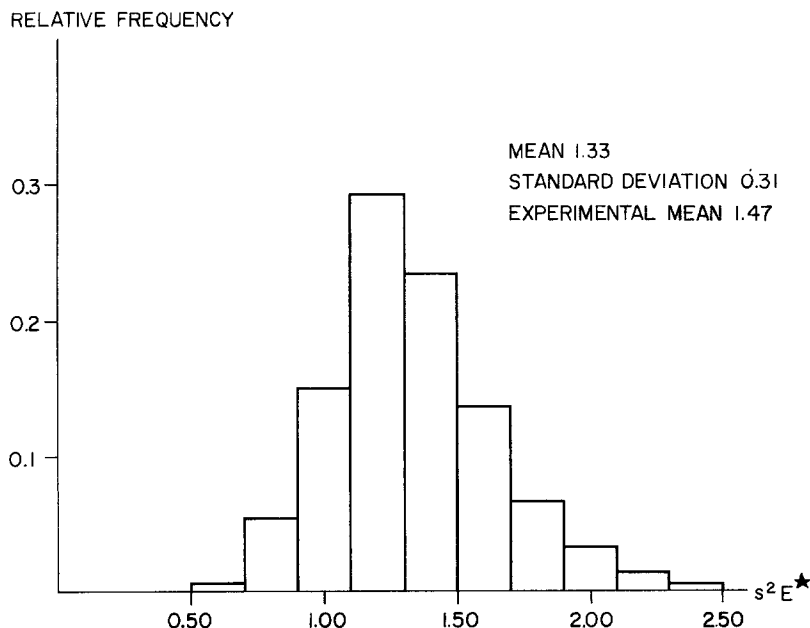


Fig. 4. Bootstrap sampling distribution for s_E^2 based on 500 bootstrap replications.

The bootstrap sampling distribution for the estimation variance (Fig. 4) shows that an approximate 95% confidence interval for s_E^2 is 0.90 to 1.90. This confidence interval was chosen to leave approximately 5% of the probability at either end.

DISCUSSION

Standard geostatistical procedures treat the autocovariance function (or, equivalently, the semivariogram) as known when, in fact, only an estimate is available. In many cases, this estimate will contain considerable uncertainty. Ignoring uncertainty leads to optimistic interval estimates of interpolated values. The bootstrap is a procedure for incorporating an estimate of uncertainty into estimation variances. The uncertainty so incorporated is that associated with keeping sample locations fixed and considering repeated realizations of the random process. In some cases, this uncertainty is not as interesting as that associated with keeping the realization fixed and considering repeated sampling. For this reason, the bootstrap is best suited to cases involving independent realizations or in which size of the unique realization is large relative to range of autocorrelation. When neither of these situations pertains, the bootstrap sample will suffer for not being conditional on data values. In particular, interval esti-

mates for estimation variances will be too wide. In this case, one approach to the problem is through conditional simulation.

The bootstrap procedure outlined here does not lead to any routine procedure for broadening confidence intervals around interpolated values. However, progress has been made in this direction which uses the bootstrap procedure in conjunction with a method for estimating the shape of error distributions.

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